Brazil Wildfires: Predicting Fire Strength from Weather and Terrain

Dataset background, data sources, and modeling challenges

Foundations of Neural Networks

Group Data Presentation

Daniel Wang, James Povilonis, Jose Perales

Motivation

- Like many countries, Brazil suffers from seasonal wildfires. From 2018 to 2025, satellites flagged **1.43 million active-fire hotspots** across the country.[3]
- Fires can last a long time. Drought in 2007 and 2010 "caused multiple fires to burn synchronously for **over 2 months** across tropical forests and savannas in South America."[1]
- With satellites data, we get more than just tracking wildfire spread. Each detection also comes with **fire radiative power (FRP)**, an estimate of how intensely it's burning.[2]
- A detection tells responders **where** a fire is, but not whether it will be one of the intense ones. Weather, terrain, and land cover might help with that.
- Run on forecast weather, a model like this could help decide where to send firefighters first.

Our question: How much do weather, terrain, and land cover at a detected hotspot tell us about how intense the fire is?

[1] Andela et al., "The Global Fire Atlas of individual fire size, duration, speed and direction," 2019.

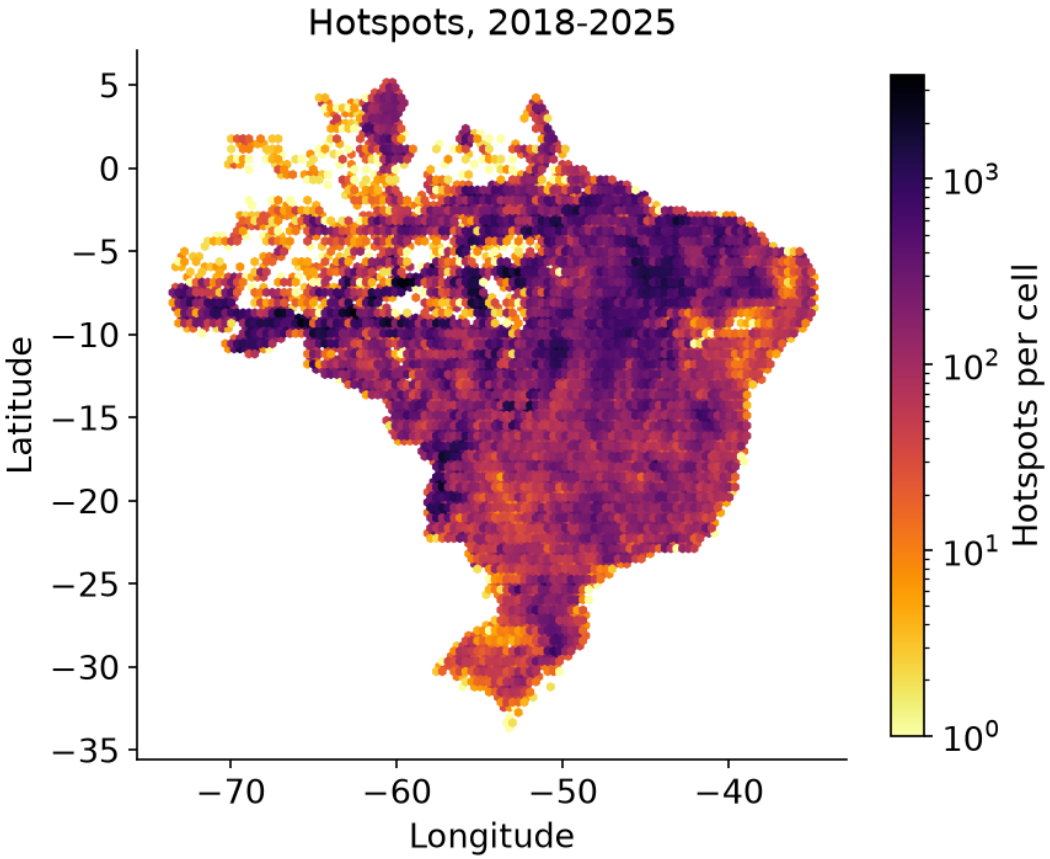
[2] Wooster et al., "Retrieval of biomass combustion rates and totals from fire radiative power observations," 2005.

[3] Costa, "Banco Histórico de Incêndios no Brasil 2018–2025," Hugging Face, 2025.

Where the dataset comes from

- Published on **Hugging Face** by Mateus P. Costa (2025) as *Banco Histórico de Incêndios no Brasil 2018–2025* (Historical Fire Database for Brazil).[3]
- It covers **all of Brazil** and its six biomes from **2018 to 2025**, with one Parquet file per year.[3]
- The card lists **tabular classification and regression** as intended tasks.[3]

Map: our EDA, all 1,430,756 hotspots.



[3] Costa, "Banco Histórico de Incêndios no Brasil 2018–2025," Hugging Face, 2025.

Background on satellite fire detection

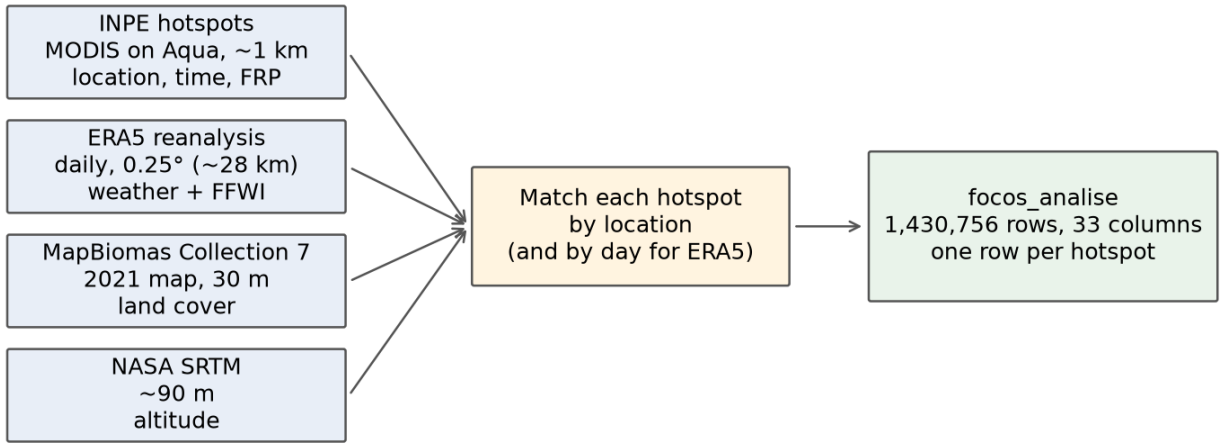
- **MODIS**, an instrument on NASA's Aqua satellite, uses thermal infrared to find unusually hot areas. Each detection covers a pixel of roughly 1 km.[4]
- A **hotspot** isn't necessarily a separate fire. A large wildfire can show up as many detections.[4]
- **FRP** (fire radiative power) measures how much energy a fire is radiating, in megawatts (MW), when the satellite passes overhead.[2]
- **FFWI** (Fosberg Fire Weather Index) combines temperature, humidity, and wind into a single measure of fire weather. It's one of the weather columns we can use as a feature.[5]

[2] Wooster et al., "Retrieval of biomass combustion rates and totals from fire radiative power observations," 2005.

[4] Giglio, Schroeder & Justice, "The collection 6 MODIS active fire detection algorithm and fire products," 2016.

[5] Fosberg, "Weather in wildland fire management: The fire weather index," 1978.

How the dataset was built



Our diagram, based on the dataset card.[3]

- The dataset starts with **hotspots reported by INPE**, Brazil's National Institute for Space Research. Each has a location, detection time, FRP, and INPE's own rain and fire-risk fields. All come from Aqua's afternoon pass (`AQUA_M-T`).[6]
- The author then matched each hotspot with **that day's weather**, plus elevation and land cover for that location.[3]

[3] Costa, "Banco Histórico de Incêndios no Brasil 2018–2025," Hugging Face, 2025.

[6] INPE, Programa Queimadas (BDQueimadas).

What does the dataset contain?

Table	What it holds	Rows[3]
focos_analise (what we use)	INPE hotspots + land cover, altitude, weather	1,430,756
focos_inpe	The same hotspots, raw from INPE	1,430,756
meteo_era5	Daily ERA5 weather grid over Brazil	84,025,032
focos_firms	NASA FIRMS VIIRS detections, **2018 only**	2,643,169

Source	Resolution	Time coverage[3]
INPE hotspots (MODIS)	~1 km	Daily, 2018–2025
ERA5 weather[7]	0.25° (~28 km)	Daily
MapBiomas land cover[8]	30 m	One map, 2021
SRTM altitude[9]	~90 m	Static

- The full dataset has about 89.5 million rows, but we're mainly using focos_analise because it joins fire intensity and environmental data.

[3] Costa, 2025. [7] Hersbach et al., "The ERA5 global reanalysis," 2020. [8] Souza et al., "Reconstructing three decades of land use and land cover changes in Brazilian biomes," 2020. [9] Farr et al., "The Shuttle Radar Topography Mission," 2007.

What does a row look like?

Column	Meaning	Example
`data_hora_gmt`	Detection time (UTC)	2023-01-02 17:11
`latitude`, `longitude`	Hotspot location	–17.53, –57.66 (Cáceres, MT)
`bioma`	Biome	Pantanal
`nome_classe_mb`	MapBiomas land cover	Campo Alagado (flooded grassland)
`altitude_m`	SRTM altitude	96 m
`temp_c`, `rh`	Daily temperature, relative humidity	29.4 °C, 73.6%
`wind_speed`, `tp_mm`	Daily wind speed, precipitation	1.9 m/s, 0.09 mm
`ffwi`	Fire weather index	0.51
`frp` (target)	Fire radiative power	82.4 MW

Example: one hotspot from 2023, showing 12 of the 33 available columns.

- The column names and categories are **in Portuguese**, so we've added English descriptions to make them easier to follow.

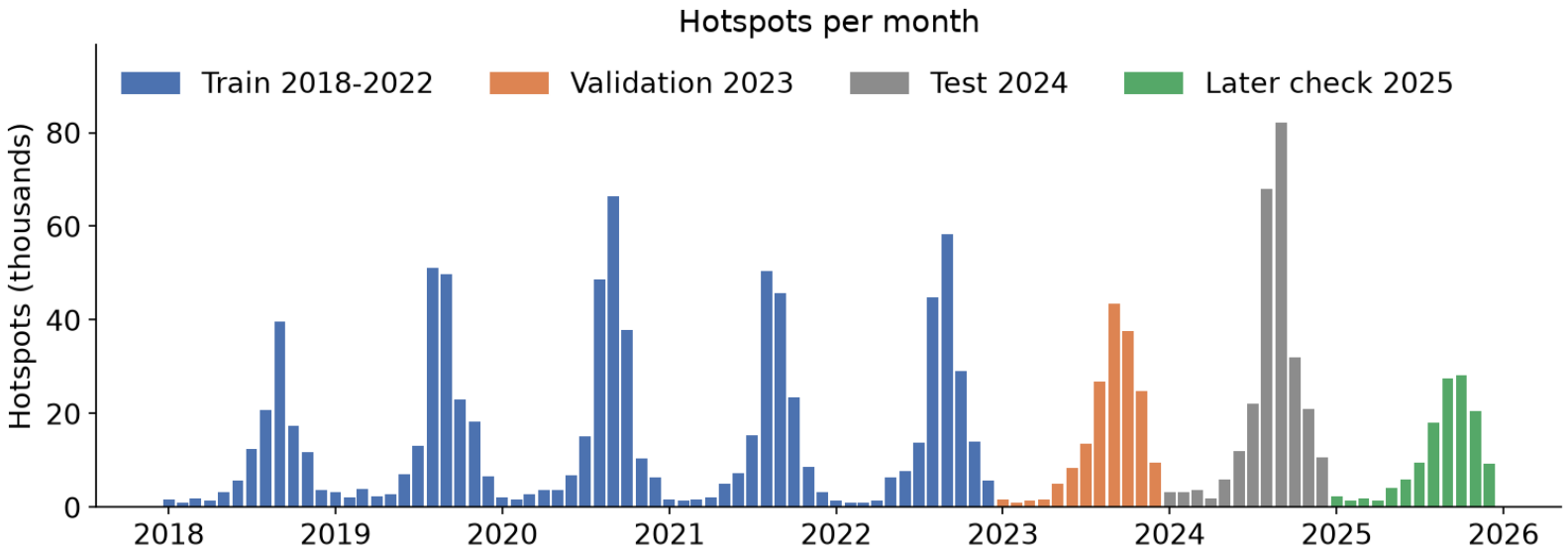
Checking the dataset before modeling

What we found	Why it matters
2,540 rows (all in 2020) have no FRP	We can't train on these without a target value
The same 2,540 rows have a geometry up to ~50 m off their coordinates	The coordinates don't quite match, and we don't know why yet
2,622 rows lack INPE's rain, dry-days, and fire-risk fields; 237 lack altitude	We'll need to decide how to handle these missing values
26 days have no hotspots, including April 1–16, 2022	This looks more like missing data than 16 days without any fires
`geometria` is written `POINT (lat lon)`, the reverse of the usual order	We need to be careful about coordinate order when using GIS tools
689 rows report over a year without rain (max 827 days)	These look suspicious and need another look

- Otherwise, our row counts **matched the dataset card**[3] and we found no duplicate IDs.

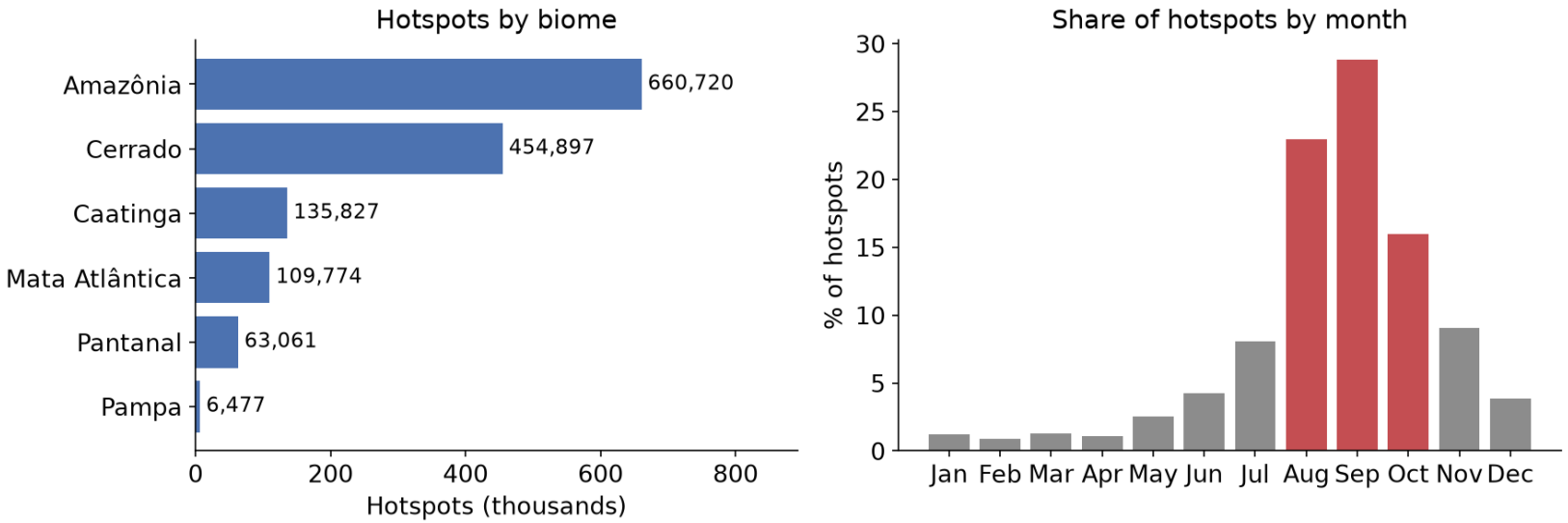
[3] Costa, "Banco Histórico de Incêndios no Brasil 2018–2025," Hugging Face, 2025.

Preparing independent evaluation data



- Rather than splitting rows at random, we're splitting **by year**. We'll train on 2018–2022, validate on 2023, and test on 2024, with 2025 held aside as another check.
- Many hotspots share the same day's weather. Across 1.43 million rows, there are only 573,960 distinct day-and-weather combinations. A random split could put **very similar rows** in both training and test.
- 2024 has the **most hotspots** of any year (266,001), so it could be a harder test year.

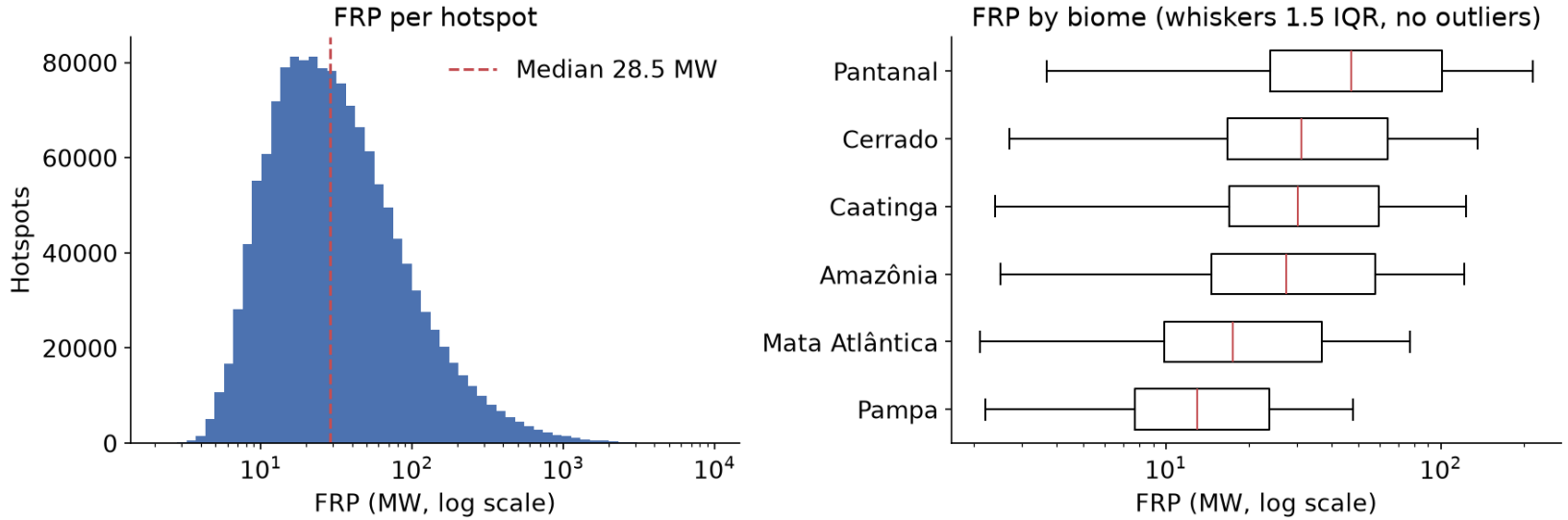
Hotspots cluster in two biomes and three months



Our EDA: all 1,430,756 hotspots. Red: August to October.

- About **78%** of detections come from Amazônia and Cerrado. Pampa has far fewer, with just 6,477 hotspots.
- Detections are also very seasonal. About **68%** happen between August and October, and September alone has nearly 29%.
- A model could do well overall but poorly in the less common biomes or months, so we should check those separately.

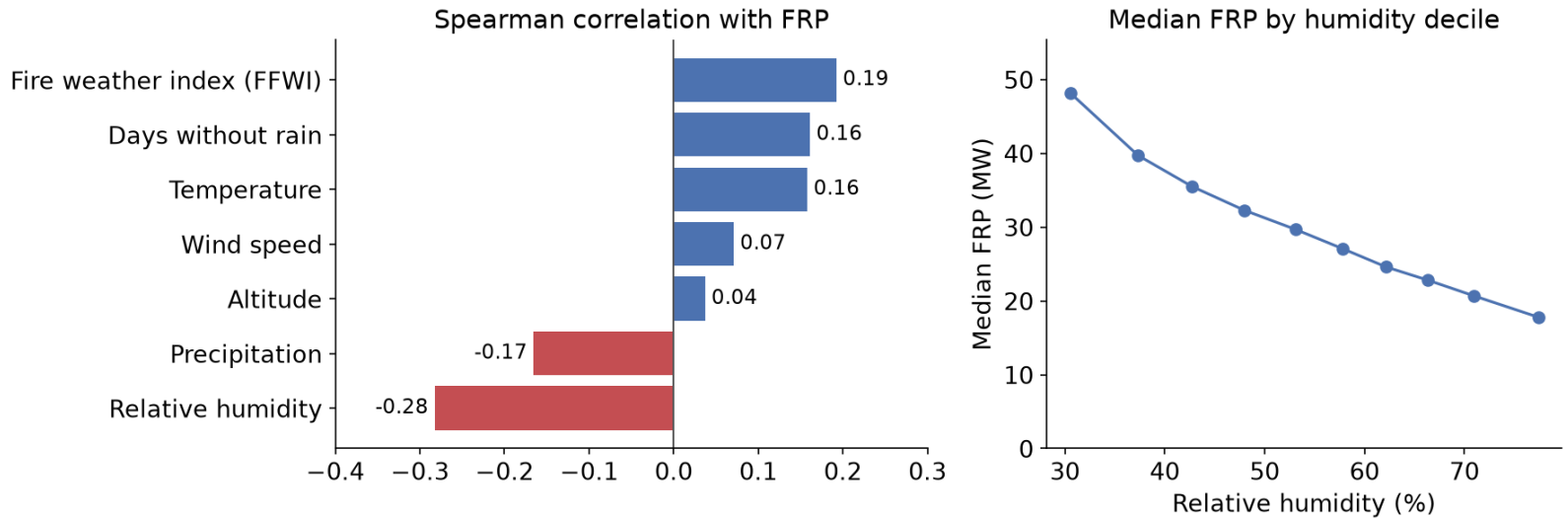
Fire intensity is heavily skewed



Our EDA: 1,428,216 hotspots with FRP. Boxes show quartiles; whiskers reach 1.5 IQR.

- Most hotspots have fairly low FRP. The median is **28.5 MW**, but a small number of very intense detections (up to 9,612 MW) pull the mean up to 63.1 MW.
- Just 1% of hotspots account for **16.9% of the total FRP**, and 5,182 detections are above 1,000 MW.
- FRP also varies by biome, from a median of 13.0 MW in Pampa to 47.1 MW in Pantanal. Since it's so skewed, we plan to model **log FRP** instead of raw FRP.

Drier weather is correlated with more intense fires



Our EDA: hotspots with FRP, all years. Rank correlations only, with no control for biome or season.

- Of the variables we looked at, **humidity** stands out most (ρ = –0.28). Median FRP falls from 48.2 MW in the driest tenth of hotspots to 17.8 MW in the most humid tenth.
- Temperature, days without rain, and FFWI have weaker positive correlations (ρ ≈ 0.16 to 0.19). Surprisingly, wind speed on its own has almost none (0.07).
- Weather changes with the seasons, so some of these relationships likely reflect **seasonal differences** rather than the effect of any one variable.

What might the data predict?

Based on what we've seen so far, these are the hypotheses we're considering to test.

Question we want to answer	Why we think it could work	What makes it difficult
H1 Can we estimate a hotspot's FRP from weather, terrain, and land cover?	We already see some links between weather and FRP	The correlations are fairly weak at $\|\rho\| \leq 0.28$
H2 Does weather tell us anything beyond location and time of year?	Weather varies even within the same season	Weather and season are closely related
H3 Do some biomes and land covers have higher FRP?	Pantanal's median FRP is 3.6× Pampa's	Some biomes, like Pampa, have far fewer examples
H4 Can we pick out unusually intense hotspots?	The highest FRP values may share recognizable conditions	Those extreme hotspots are rare

Limitations

- We only have data for fires the satellite **detected**. We can study intensity, but we can't use this table alone to predict where new fires will start.[3]
- We're relying on **one satellite's afternoon pass**. Fires can be missed if they're under cloud, burn at night, or happen between passes.[4]
- The weather is much **coarser** than the hotspots: one daily value for an area about 28 km across. The card doesn't say how ERA5's hourly values were combined.[3]
- FFWI in this dataset only ranges from **0 to 4.8**. We suspect it isn't on Fosberg's original scale, but we can't confirm that from the card.[3] [5]
- The land cover map is from **2021**, while the hotspots span 2018–2025. Some places may have changed in between.

[3] Costa, 2025. [4] Giglio, Schroeder & Justice, 2016. [5] Fosberg, 1978.

Why the dataset is useful

- The weather, terrain, and land cover are **already matched** to the fire detections, so we don't have to build those joins ourselves.[3]
- With **eight years** of data, we can test on different years instead of relying on a random split.
- The **full ERA5 weather grid** (84 million rows) is included too. With more work, it could support a different project on where fires occur, not just how intense they are.[3]
- Products like the Global Fire Atlas focus on burned area and fire duration.[1] This one gives us an **FRP value for each detection**, along with the joined weather data.

[1] Andela et al., "The Global Fire Atlas of individual fire size, duration, speed and direction," 2019.

[3] Costa, "Banco Histórico de Incêndios no Brasil 2018–2025," Hugging Face, 2025.

Conclusions

- This dataset is a good fit for our question. It has **eight years** of hotspots with weather, terrain, and land cover already attached.
- Hotspots aren't spread evenly. Most fall in **two biomes and a few months**, and a small number have very high FRP.
- Weather seems related to FRP, but the correlations **aren't very strong**. We still need to see whether they hold up once we account for location and season.
- There are some **data quality issues** to deal with first, like missing values, suspicious records, and gaps in coverage.
- **Our question going into modeling:** How much do weather, terrain, and land cover at a detected hotspot tell us about how intense the fire is?

References

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